

Andrea T. Ricolfi

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EMPLOYMENT HISTORY & EDUCATION

Abilitazione Nazionale I fascia	10 July 2025 –
Assistant Professor (rtd-B) at SISSA	July 2022 – Sept 2024
Assistant Professor (rtd-B) at Università di Bologna	Sept 2021 – June 2022
Postdoc at SISSA, Trieste (<i>SISSA Mathematical Fellowship</i>)	Nov 2018 – Sept 2021
Postdoc at Max-Planck Institut für Mathematik, Bonn	Nov 2017 – Oct 2018
PhD in Mathematics at University of Stavanger (UiS Norway)	Sept 2013 – Oct 2017
Thesis (10th Nov 2017): Local Donaldson–Thomas invariants and their refinements	
Trial Lecture: <i>Symmetric obstruction theories and Joyce’s perverse sheaves</i>	
M.S. in Mathematics (ALGANT Program: Padova & Bordeaux 1)	Oct 2010 – July 2012

VISITS AND SCHOLARSHIPS

Imperial College London Visiting PhD (P.I. Prof. Richard Thomas)	Feb 2015 – June 2015
University of Copenhagen 4 short term visits (P.I. Prof. Lars H. Halle)	2015 – 2017
SISSA One month Research Scholarship	June 2013

RESEARCH INTERESTS

Enumerative geometry of moduli spaces of sheaves • Hilbert and Quot schemes, Donaldson–Thomas invariants, virtual classes, virtual localisation • Moduli stacks, quiver representations, d-critical loci, derived algebraic geometry • Grothendieck rings of varieties • Hall algebras • Moduli spaces of curves, compactified Jacobians

SUPERVISION

PhD students	Defense date
◦ Anna Sicolo (SISSA), co-supervised with Prof. F. Carocci.	ongoing
◦ Federico Candeletti (SISSA), co-supervised with Prof. D. Lewanski.	ongoing
◦ Luca Morstabilini (SISSA), co-supervised with Prof. F. Sala.	ongoing
◦ Nicolò Bignami (SISSA), co-supervised with Prof. A. Marian.	ongoing
• Valentina Bais (SISSA), co-supervised with Prof. D. Zuddas.	16 Sep 2026
• Solomiya Mizyuk (SISSA), co-supervised with Prof. B. Fantechi.	14 May 2026
• Michele Graffeo (SISSA), co-supervised with Prof. U. Bruzzo.	25 Nov 2022
Master students	Defense date
• Lorenzo Palcic (Università di Trieste).	17 Jul 2025
• Riccardo Redigolo (Università di Trieste), co-supervised with Prof. B. Fantechi.	12 Jul 2024

CONFERENCE ORGANISATION

Moduli Spaces in (Super)Geometry and Mathematical Physics (João Pessoa, Brazil)	6–10 Oct 2025
The geometry of Hilbert schemes of points (Levico Terme)	6–10 May 2024
Quiver Representations, Quiver Varieties and Combinatorics (Bologna)	22–26 May 2023
Refined invariants in moduli theory (SISSA and UNITS)	2–5 May 2023
Moduli spaces: theory and coding (Les Diablerets)	27 Feb – 3 Mar 2023
Derived Categories and Moduli Spaces , local organiser (Stavanger)	8–10 Sept 2015

Published papers

30. *Deformation theory for a morphism in the derived category with fixed lift of the codomain*, with P. Belmans, W. Lowen and S. Okawa.
Int. Math. Res. Not. IMRN, Vol. **2026**, Issue 14, rnag162 (2026).
29. *Invariants of nested Hilbert and Quot schemes on surfaces*, with N. Fasola, M. Graffeo, D. Lewański, 2025.
J. Math. Pures Appl., Vol. **216**, 103973 (2026).
28. *The motive of the Hilbert scheme of points in all dimensions*, with M. Graffeo, S. Monavari, R. Moschetti.
Proc. Lond. Math. Soc, Vol. **132**, No. 3, e70140, pp. 1–44 (2026).
27. *Indecomposability of derived categories in families*, with F. Bastianelli, P. Belmans and S. Okawa.
J. Geom. Phys., Vol. **217**, 105600 (2025).
26. *On the stack of 0-dimensional coherent sheaves: motivic aspects*, with B. Fantechi.
Bull. Lond. Math. Soc., Vol. **57**, No. 6, 1607–1649 (2025).
25. *Hyperquot schemes on curves: virtual class and motivic invariants*, with S. Monavari.
Math. Ann., Vol. **392**, No. 2, 1665–1709 (2025).
24. *The geometry of double nested Hilbert schemes of points on curves*, with M. Graffeo, P. Lella, S. Monavari and A. Sammartano.
Trans. Amer. Math. Soc., Vol. **378** (2025), 6013–6047.
23. *On the stack of 0-dimensional coherent sheaves: structural aspects*, with B. Fantechi.
In: *Moduli, Motives and Bundles: New Trends in Algebraic Geometry*. London Mathematical Society Lecture Note Series **499**. Cambridge University Press (2025) 1–32.
22. *A sign that used to annoy me, and still does*.
J. Geom. Phys., Vol. **125**, 105032 (2024).
21. *The d -critical structure on the Quot scheme of points of a Calabi–Yau 3-fold*, with M. Savvas.
Commun. Contemp. Math., Vol. **415**, No. 8, 1–45 (2024).
20. *On the Behrend function and the blowup of some fat points*, with M. Graffeo.
Adv. Math., Vol. **415** (2023), 108896.
19. *Hilbert squares of degeneracy loci*, with E. Fatighenti, F. Meazzini, G. Mongardi.
Rend. Circ. Mat. Palermo (2), **72**, 3153–3183 (2023).
18. *On the motive of the nested Quot scheme of points on a curve*, with S. Monavari.
Journal of Algebra, Vol. **610**, 99–118 (2022).
17. *Sur la lissité du schéma Quot ponctuel emboîté*, with S. Monavari.
Canad. Math. Bull., Vol. **66**, Issue 1, March 2023, pp. 178–184.
16. *Framed sheaves on projective space and Quot schemes*, with A. Cazzaniga.
Math. Z. **300**, 745–760 (2022).
15. *Framed motivic Donaldson–Thomas invariants of small crepant resolutions*, with A. Cazzaniga.
Math. Nachr. Vol. **295**, Issue 6 (2022), 1096–1112.
14. *Higher rank motivic Donaldson–Thomas invariants of $\mathbb{A}^3$ via wall-crossing, and asymptotics*, with A. Cazzaniga and D. Ralaivaosaona.
Math. Proc. Cambridge Philos. Soc., Vol. **174**, Issue 1, January 2023, pp. 97–122.
13. *Higher rank K -theoretic Donaldson–Thomas theory of points*, with N. Fasola and S. Monavari.
Forum Math. Sigma, Vol. **9** E15 (2021) 1–51.
12. *The equivariant Atiyah class*.
C. R. Math. Acad. Sci. Paris., Vol. **359**, Issue 3 (2021), 257–282.
11. *On the motive of the Quot scheme of finite quotients of a locally free sheaf*.
J. Math. Pures Appl., **144** (2020), 50–68.
10. *Virtual classes and virtual motives of Quot schemes on threefolds*.
Adv. Math., Vol. **369**, 107182 (2020).
9. *The local motivic DT/PT correspondence*, with B. Davison.
J. Lond. Math. Soc., Vol. **104**, Issue 3, 1384–1432 (2021).
8. *Virtual counts on Quot schemes and the higher rank local DT/PT correspondence*, with S. Beentjes.
Math. Res. Lett., Vol. **28**, no. 4 (2021), 967–1032.
7. *Pullbacks of universal Brill–Noether classes via Abel–Jacobi morphisms*, with N. Pagani and J. van Zelm.
Math. Nachr., Vol. **293**, Issue 11 (2020), 2187–2207.
6. *The Hilbert scheme of hyperelliptic Jacobians and moduli of Picard sheaves*.
Algebra Number Theory **14**, No. 6 (2020), 1381–1397.
5. *Jet bundles on Gorenstein curves and applications*, with L. Gatto.

- J. Singul. Volume **21** (2020), 50–83.
4. [The DT/PT correspondence for smooth curves](#).
Math. Z. **290** (2018), no. 1-2, 699–710.
 3. [On coherent sheaves of small length on the affine plane](#), with R. Moschetti.
J. Algebra, **516** (2018), pp. 471–489.
 2. [Local contributions to Donaldson–Thomas invariants](#).
Int. Math. Res. Not. IMRN, **2018** (2018), no. 19, 5995–6025.
 1. [The Euler characteristic of the generalized Kummer scheme of an Abelian threefold](#), with M. Gulbrandsen.
Geom. Dedicata, **182** (2016), Issue 1, pp. 73–79.

Preprints

1. [Motivic and cohomological stabilisation of the Quot scheme of points](#), with M. Graffeo, S. Monavari, R. Moschetti, 2026.
2. [Enumeration of partitions via socle reduction](#), with M. Graffeo, S. Monavari, R. Moschetti, 2025.
3. [Derived hyperquot schemes](#), with S. Monavari, E. Pavia, 2024.
4. [Motivic classes of noncommutative Quot schemes](#), 2023.
5. [Moduli spaces of semiorthogonal decompositions in families](#), with P. Belmans and S. Okawa, 2020.

Books

1. [An Invitation to Modern Enumerative Geometry](#), Springer, SISSA Lecture Series, Vol. 3 (2022).

TALKS AT INTERNATIONAL CONFERENCES AND WORKSHOPS

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| – HOPS – HOlomorphic, Projective and Symplectic Geometry in Bari
<i>Rank 2 sheaves on $\mathbb{P}^3$, Quot schemes and wall-crossing</i> | Sep 2026 |
| – Algebraic Geometry in the Daylight
<i>Rank 2 sheaves on $\mathbb{P}^3$, Quot schemes and wall-crossing</i> | Aug 2026 |
| – Modern Methods in Moduli (Luxembourg)
<i>The motive of the Hilbert scheme of points</i> | Nov 2024 |
| – From Schubert Calculus to Representation Theory (Aracaju, Brazil)
<i>The motive of the Hilbert scheme of points</i> | Sept 2024 |
| – Categorified Enumerative Geometry and Representation Theory (EPFL, Lausanne)
<i>Structures on the Quot scheme of points of a Calabi–Yau 3-fold</i> | Sept 2023 |
| – Higher Structures in Geometry and Mathematical Physics (IHP, Paris - online)
<i>d-critical structure(s) on the Quot scheme of points of a Calabi–Yau 3-fold</i> | June 2023 |
| – IV Congresso Brasileiro de Jovens Pesquisadores (João Pessoa, Brazil)
<i>Higher rank K-theoretic Donaldson–Thomas theory</i> | Oct 2022 |
| – Young Researchers Meeting in Algebra and Geometry 2022 (SISSA, Trieste)
<i>A tale of two d-critical structures</i> | Sept 2022 |
| – Bandoleros 2022 (Ankara, online)
<i>A tale of two d-critical structures</i> | May 2022 |
| – Grothendieck ring and Derived category: a gathering (Turin)
<i>A motivic wall-crossing formula</i> | Apr 2022 |
| – Bandoleros 2021 (Campinas, online)
<i>Virtual invariants of Quot schemes on 3-folds</i> | Feb 2021 |
| – HMI Workshop on Gauge theory and virtual invariants (Dublin)
<i>Virtual classes and virtual motives of Quot schemes on 3-folds</i> | May 2019 |
| – Workshop in Algebraic Geometry (Milan)
<i>A higher rank local DT/PT correspondence</i> | Dec 2018 |
| – Algebraic Geometry and Foliations: Vainsencher 70 (Belo Horizonte, Brazil)
<i>A component of the Hilbert scheme of hyperelliptic Jacobians</i> | Nov 2018 |
| – Motives of Calabi–Yau manifolds (Kraków)
<i>A motivic wall-crossing formula for sheaves on 3-folds</i> | May 2018 |
| – A Fall Meeting in Algebraic Geometry and related topics (Turin)
<i>The DT/PT correspondence for smooth curves</i> | Oct 2017 |

– National Algebra meeting (Oslo) <i>Local contributions to DT invariants</i>	Nov 2016
– National Algebra meeting (Oslo) <i>Critical loci and their virtual motives</i>	Nov 2015
– Moduli Spaces and Derived Categories (Warwick) <i>Partitions and generalized Kummer varieties</i>	Feb 2015
– GAel XXII (SISSA) <i>Motivic DT invariants</i>	June 2014
– National Algebra meeting (Oslo) <i>Limits of special Weierstrass points</i>	Nov 2013

SELECTED SEMINAR TALKS

– <i>Moduli of semiorthogonal decompositions</i> (Tor Vergata, Rome)	June 2026
– <i>Moduli of semiorthogonal decompositions</i> (University of Strasbourg)	May 2026
– <i>Moduli of semiorthogonal decompositions</i> (University of Padua)	Dec 2025
– <i>The motive of the Hilbert scheme of points</i> (University of Utrecht)	Apr 2024
– <i>Geometry of Hilbert schemes, and the two numbers $+1, -1$</i> (Politecnico di Milano)	Apr 2023
– <i>Quot schemes and their d-critical structure(s)</i> (Firenze)	Mar 2023
– <i>Enumerative invariants of Quot schemes and their virtual refinements</i> (ICTP)	Feb 2023
– <i>Quot schemes and their d-critical structure(s)</i> (Pisa)	Nov 2022
– <i>Quot schemes and their d-critical structure(s)</i> (Bonn)	Nov 2022
– <i>K-theoretic sheaf counting</i> (Genova)	May 2022
– <i>A motivic DT/PT correspondence</i> (EPFL, Lausanne)	May 2022
– <i>Refined invariants of moduli spaces</i> (Math. Colloquium , João Pessoa, Brazil)	Apr 2022
– <i>Refined sheaf counting</i> (Trento)	Feb 2022
– <i>Sheaf counting and Quot schemes</i> (Milan)	Nov 2021
– <i>d-critical structure(s) on the Quot scheme of points on a 3-fold</i> (CMSA Harvard)	Oct 2021
– <i>The d-critical structure on the Quot scheme of points on a 3-fold</i> (SISSA)	May 2021
– <i>Refinements of higher rank DT invariants</i> (KIAS Seoul, remote)	May 2021
– <i>Higher rank motivic DT invariants</i> (SISSA)	Feb 2021
– <i>Higher rank K-theoretic Donaldson–Thomas theory of points</i> (Kansas, remote)	Oct 2020
– <i>Higher rank K-theoretic Donaldson–Thomas theory of points</i> (Bologna)	Oct 2020
– <i>A moduli space of semiorthogonal decompositions</i> (Rutgers New Jersey, remote)	Sept 2020
– <i>Higher rank K-theoretic Donaldson–Thomas theory of points</i> (UCSD, remote)	Apr 2020
– <i>Moduli of semiorthogonal decompositions</i> (Stavanger)	Nov 2019
– <i>A motivic DT/PT correspondence via Quot schemes</i> (Oxford)	Nov 2019
– <i>Virtual invariants of Quot schemes on 3-folds</i> (Copenhagen)	May 2019
– <i>A component of the Hilbert scheme of hyperelliptic Jacobians</i> (Rome)	Apr 2019
– <i>Curve counting via Quot schemes</i> (Utrecht University)	Dec 2018
– <i>Le schéma de Hilbert d'une Jacobienne hyperelliptique</i> (Nancy)	Oct 2018
– <i>The DT/PT correspondence for smooth curves</i> (University of Edinburgh)	Jan 2018
– <i>The DT/PT correspondence for smooth curves</i> (KTH, Stockholm)	Nov 2017
– <i>Counting rational curves on toric 3-folds</i> (Copenhagen)	Feb 2016
– <i>Families of Abel–Jacobi curves</i> (Turin)	Dec 2015
– <i>Curve counting on threefolds</i> (Bergen)	Oct 2015
– <i>Introduction to Motivic Integration</i> (Imperial College London)	Apr 2015
– <i>Refined curve counting on Calabi–Yau 3-folds</i> (KU Leuven)	Mar 2015
– <i>Localisation in Donaldson–Thomas theory</i> (UCL, London)	Feb 2015
– <i>A Hamilton’s Principle in Algebraic Geometry</i> (Turin)	Dec 2014
– <i>Curve Counting and Box Counting</i> (Turin)	June 2014
– <i>Curve Counting Invariants and Euler Characteristics</i> (Bergen)	Feb 2014

TEACHING

<i>Enumerative Geometry</i> (20h, SISSA PhD course)	Spring 2027
<i>Algebraic Geometry</i> (60h, 2 nd Year Master Università di Trieste and SISSA PhD)	Fall 2026

<i>Enumerative Geometry</i> (30h, SISSA PhD course)	Spring 2026
<i>Algebraic Geometry</i> (50h, 2 nd Year Master Università di Trieste and SISSA PhD)	Fall 2024
<i>Algebraic Geometry</i> (60h, 2 nd Year Master Università di Trieste and SISSA PhD)	Fall 2023
<i>Algebraic Geometry</i> (60h, 2 nd Year Master Università di Trieste and SISSA PhD)	Fall 2022
<i>Geometria e Algebra T</i> (60h, Ingegneria Chimica e Biochimica, Bologna)	Fall 2021
<i>Localisation in Enumerative Geometry</i> (20h, SISSA PhD course)	Spring 2021
<i>Techniques in Enumerative Geometry</i> (20h, SISSA PhD course)	Fall 2019
<i>Algebraic Geometry MAT630</i> (40h, Master course, University of Stavanger)	Spring 2017
<i>Mathematical Methods 2 MAT200</i> (T.A. Bachelor, University of Stavanger)	Spring 2016
<i>Linear Algebra MAT110</i> (T.A. Bachelor, University of Stavanger)	Fall 2015
<i>Discrete Mathematics MAT120</i> (40h, Bachelor course, University of Stavanger)	Fall 2014
<i>Geometria e Algebra Lineare</i> (T.A. Bachelor, Politecnico di Torino)	Spring 2013

OTHER TASKS

Member of <i>Commissione Didattica</i> (SISSA)	1 July 2025–
Member of <i>Collegio dei docenti di Dottorato</i> (SISSA)	1 July 2022–
Organiser of SISSA AG Seminar , TRINO and SISSA-ICTP Quot Seminar	1 Oct 2022–